

# Lyapunov-Based Adaptive Observers and Continuous-Time Recurrent Neural Networks for Uncertain Nonlinear Systems: A Theoretical Synthesis

## 1. Architectural Lineage and the Neural Observer Paradigm

State estimation in uncertain nonlinear dynamical systems represents a fundamental challenge in modern control engineering, robotics, and cyber-physical systems<sup>1</sup>. While linear systems admit dual separation principles that decouple state reconstruction from state-feedback regulation, nonlinear systems generally lack universal separation theorems<sup>1</sup>. Historically, nonlinear state observation relied on three primary paradigms: high-gain observers (HGO), sliding-mode observers (SMO), and Extended Kalman Filters (EKF)<sup>1</sup>. Each of these classical methodologies exhibits structural trade-offs<sup>1</sup>. High-gain observers reconstruct unmeasured states rapidly but suffer from severe sensitivity to high-frequency sensor noise and the peaking phenomenon, which can destabilize actuators during transients<sup>1</sup>. Sliding-mode observers provide disturbance rejection but introduce high-frequency chattering that excites unmodeled high-frequency plant dynamics<sup>4</sup>. Meanwhile, Kalman-based filtering relies on localized Taylor-series linearizations that lack deterministic stability guarantees in the presence of unmodeled nonlinear friction, hysteresis, or non-Gaussian disturbances<sup>3</sup>.

To overcome these structural limitations, adaptive observers utilizing artificial neural networks emerged as a powerful framework capable of approximating unknown nonlinearities online<sup>1</sup>. Early neural observer designs incorporated static feedforward architectures, such as single-layer multilayer perceptrons or radial basis function networks<sup>4</sup>. While supported by universal approximation theorems on compact domains, static networks map instantaneous inputs directly to instantaneous outputs without maintaining temporal state memory<sup>7</sup>. As a consequence, static neural networks struggle to model complex dynamical phenomena characterized by internal states, dynamic friction, or high-order differential couplings<sup>8</sup>. This limitation catalyzed a paradigm shift toward continuous-time dynamic and recurrent neural networks (RNNs)<sup>1</sup>. By integrating internal recurrent states, continuous-time recurrent networks serve as dynamic filters that emulate the differential structure of the underlying physical plant, thereby capturing time-varying and cumulative dynamic effects<sup>1</sup>.

A seminal research lineage in this domain stems from Prof. Warren E. Dixon and his collaborators at the Nonlinear Controls and Robotics (NCR) Laboratory at the University of Florida<sup>11</sup>. The NCR laboratory pioneered methodologies that rigorously interface non-convex machine learning models with continuous and nonsmooth Lyapunov stability theory<sup>13</sup>. Rather

than relying on heuristic optimization schemes or offline gradient backpropagation through time (BPTT)—which offer no closed-loop stability guarantees—the Dixon group formulated continuous-time adaptive update laws derived analytically from Lyapunov candidates<sup>13</sup>. This theoretical progression traces from shallow dynamic neural networks to deep recurrent architectures, ultimately yielding continuous-time Lyapunov-based Long Short-Term Memory (Lb-LSTM) networks and physics-informed models that structurally enforce conservation laws within the estimator<sup>1</sup>.

| Investigation                              | Core Architecture                                                       | Primary Theoretical Innovation                                                                                   | Convergence Outcome                                                   | Benchmark Application                               |
|--------------------------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------|
| <b>Dinh et al. (2014)</b><br>[cite: 1]     | Shallow Dynamic Neural Network (DNN) + Robust Sliding Mode <sup>1</sup> | Second-order auxiliary dynamic filter eliminating unmeasurable velocity dependencies <sup>1</sup>                | Asymptotic state error convergence via Lyapunov analysis <sup>1</sup> | Two-link planar robot manipulator <sup>1</sup>      |
| <b>Griffis et al. (2023)</b><br>[cite: 10] | Deep Recurrent Neural Network (DRNN) <sup>10</sup>                      | Real-time adaptation across cascaded recurrent layers via analytical chain-rule Jacobians <sup>8</sup>           | Semi-global uniform ultimate boundedness (UUB) <sup>2</sup>           | Multi-state uncertain nonlinear plants <sup>8</sup> |
| <b>Griffis et al. (2024)</b><br>[cite: 15] | Continuous-Time Lyapunov-Based LSTM (Lb-LSTM) <sup>15</sup>             | Euler-discretized continuous gating dynamics and multi-gate analytical Jacobian sensitivity updates <sup>9</sup> | Semi-global asymptotic state reconstruction <sup>15</sup>             | Euler-Lagrangian mechanical systems <sup>20</sup>   |
| <b>Hart et al. (2024)</b>                  | Physics-Informed                                                        | Modular neural parametrization                                                                                   | Asymptotic state tracking                                             | Fully coupled Euler-Lagrangian                      |

|            |                                             |                                                                                  |                                                     |                                        |
|------------|---------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------|----------------------------------------|
| [cite: 16] | Lyapunov-Based LSTM (PI-LSTM) <sup>16</sup> | n enforcing positive-definite inertia and skew-symmetry invariants <sup>16</sup> | and parameter identification under PE <sup>16</sup> | e dynamics with friction <sup>21</sup> |
|------------|---------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------|----------------------------------------|

## 2. Baseline Adaptive Dynamic Neural Network Observers

### Mathematical Formulation and System Class

The foundation of the modern continuous-time neural observer lineage was established by Dinh, Kamalapurkar, Bhasin, and Dixon (2014) for a broad class of  $N^{\text{th}}$ -order uncertain, control-affine nonlinear dynamical systems formulated in observer canonical form<sup>1</sup>:

$$\begin{aligned}
 \dot{x}_1 &= x_2 \\
 \dot{x}_2 &= x_3 \\
 &\vdots \\
 \dot{x}_{N-1} &= x_N \\
 \dot{x}_N &= f(x) + G(x)u + d(t) \\
 y &= x_1
 \end{aligned} \tag{1}$$

where  $x = [x_1, x_2, \dots, x_N]^T \in \mathbb{R}^N$  represents the state vector,  $u \in \mathbb{R}^m$  is the bounded control input,  $d(t) \in \mathbb{R}$  represents an unknown, bounded, differentiable exogenous disturbance with bounded time derivative, and  $f : \mathbb{R}^N \rightarrow \mathbb{R}$  and  $G : \mathbb{R}^N \rightarrow \mathbb{R}^{1 \times m}$  are unknown, continuously differentiable nonlinear functions<sup>1</sup>. The fundamental sensing constraint dictates that only the output  $y$  and its first  $N - 1$  time derivatives—namely the sub-vector  $[x_1, x_2, \dots, x_{N-1}]^T$ —are directly measurable, whereas the highest-order state  $x_N$  (corresponding to velocity or generalized momentum in physical systems) remains unmeasurable<sup>1</sup>.

To observe the unmeasured state  $x_N$  and simultaneously identify the plant dynamics online, a multi-layer dynamic neural network (MLDNN) observer was introduced<sup>1</sup>:

$$\begin{aligned}\dot{\hat{x}}_1 &= \hat{x}_2 \\ &\vdots \\ \dot{\hat{x}}_{N-1} &= \hat{x}_N \\ \dot{\hat{x}}_N &= \hat{W}_f^T \sigma_f(\hat{x}) + \sum_{i=1}^m \hat{W}_{gi}^T \sigma_{gi}(\hat{x}) u_i + v(t)\end{aligned}\tag{2}$$

where  $\hat{x} = [\hat{x}_1, \dots, \hat{x}_N]^T \in \mathbb{R}^N$  represents the state estimate vector,  $\hat{W}_f, \hat{W}_{gi} \in \mathbb{R}^L$  denote dynamically updated weight vectors,  $\sigma_f, \sigma_{gi} : \mathbb{R}^N \rightarrow \mathbb{R}^L$  denote activation functions evaluated using the observer states (creating continuous recurrent internal feedback), and  $v(t) \in \mathbb{R}$  is an auxiliary robustifying feedback term designed to compensate for functional reconstruction inaccuracies and exogenous disturbances<sup>1</sup>.

### The Unmeasurable Velocity Error Dilemma

To analyze error convergence, state estimation errors are recursively defined as<sup>1</sup>:

$$\tilde{x}_1 = x_1 - \hat{x}_1, \quad \tilde{x}_2 = \dot{\tilde{x}}_1 + \alpha_1 \tilde{x}_1, \quad \dots, \quad \tilde{x}_j = \dot{\tilde{x}}_{j-1} + \alpha_{j-1} \tilde{x}_{j-1} + \tilde{x}_j\tag{3}$$

for  $j = 3, \dots, N-1$ , where  $\alpha_i > 0$  are design constants<sup>1</sup>. The filtered estimation error is defined as<sup>1</sup>:

$$r = \dot{\tilde{x}}_{N-1} + \alpha \tilde{x}_{N-1} + \eta\tag{4}$$

where  $\alpha > 0$  is a feedback gain and  $\eta \in \mathbb{R}$  is an auxiliary filter signal<sup>1</sup>.

The structural obstacle in high-order nonlinear observer design centers on the term

$\dot{\tilde{x}}_{N-1} = x_N - \hat{x}_N = \tilde{x}_N$ <sup>1</sup>. Because  $x_N$  is unmeasurable, the error signal  $\tilde{x}_N$  is

fundamentally unavailable for feedback<sup>1</sup>. In classical sliding-mode observer design, establishing asymptotic stability typically necessitates injecting discontinuous feedback proportional to  $\text{sgn}(r)$  or high-gain feedback proportional to  $r^1$ . However, evaluating  $r$  requires direct knowledge of  $\tilde{x}^N$ , rendering direct feedback mathematically impossible<sup>1</sup>. Generating an ad-hoc proxy for  $x^N$  via numerical differentiation (dirty derivatives) of  $x^{N-1}$  severely amplifies measurement noise and introduces phase lag, both of which destabilize high-order adaptive loops<sup>1</sup>.

## The Auxiliary Dynamic Filter Formulation

Dinh et al. resolved this dilemma by formulating a second-order auxiliary dynamic filter that uses only the measurable error signal  $\tilde{x}^{N-1}$  as an input, generating dynamic variables that inject stabilizing negative feedback into the unmeasurable filtered error  $r^1$ . The auxiliary dynamic filter is defined by the state equations<sup>1</sup>:

$$\begin{aligned}\eta &= p - (k + \alpha)\tilde{x}_{N-1} \\ \dot{p} &= -(k + 2\alpha)p - \tilde{x}_f + [(k + \alpha)^2 + 1] \tilde{x}_{N-1} \\ \dot{\tilde{x}}_f &= p - \alpha\tilde{x}_f - (k + \alpha)\tilde{x}_{N-1}\end{aligned}\tag{5}$$

with the boundary initializations  $p(0) = (k + \alpha)\tilde{x}_{N-1}(0)$  and  $\tilde{x}_f(0) = 0$ , where  $k, \alpha \in \mathbb{R}_{>0}$  denote constant design gains<sup>1</sup>.

To trace how this filter algebraically eliminates unmeasurable velocity dependencies, consider the differential relationships connecting the internal filter states. From (5), the state  $\eta(t)$  satisfies the kinematic relation<sup>1</sup>:

$$\eta = \dot{\tilde{x}}_f + \alpha\tilde{x}_f\tag{6}$$

Taking the formal time derivative of  $\eta(t)$  along the trajectories of (5) produces<sup>1</sup>:

$$\dot{\eta} = \dot{p} - (k + \alpha)\dot{\tilde{x}}_{N-1}\tag{7}$$

Substituting the explicit differential equation for  $\dot{p}$  from (5) directly into (7) yields<sup>1</sup>:

$$\dot{\eta} = -(k + 2\alpha)p - \tilde{x}_f + [(k + \alpha)^2 + 1] \tilde{x}_{N-1} - (k + \alpha)\dot{\tilde{x}}_{N-1} \quad (8)$$

By utilizing the algebraic relationship  $p = \eta + (k + \alpha)\tilde{x}_{N-1}$  from (5), equation (8) can be reorganized as<sup>1</sup>:

$$\begin{aligned} \dot{\eta} &= -(k + 2\alpha)(\eta + (k + \alpha)\tilde{x}_{N-1}) - \tilde{x}_f + [(k + \alpha)^2 + 1] \tilde{x}_{N-1} - (k + \alpha)\dot{\tilde{x}}_{N-1} \\ &= -(k + \alpha)(\dot{\tilde{x}}_{N-1} + \alpha\tilde{x}_{N-1} + \eta) - \alpha\eta + \tilde{x}_{N-1} - \tilde{x}_f \end{aligned} \quad (9)$$

Recognizing the term within the parentheses as the exact definition of the unmeasurable filtered error  $r$  from (4), equation (9) simplifies to<sup>1</sup>:

$$\dot{\eta} = -(k + \alpha)r - \alpha\eta + \tilde{x}_{N-1} - \tilde{x}_f \quad (10)$$

Equation (10) reveals the core mechanism of the filter: although  $r(t)$  contains the unmeasurable velocity error  $\tilde{x}_N$  and cannot be measured for direct feedback injection, the internal dynamic evolution of  $\dot{\eta}$  generates an exact negative feedback component  $-(k + \alpha)r$  driven solely by measurable signals<sup>1</sup>.

The robustifying observer feedback term  $v(t)$  in (2) is subsequently synthesized entirely from measurable quantities<sup>1</sup>:

$$v(t) = -[\gamma(k + \alpha) + 2\alpha]\eta + (\gamma - \alpha^2)\tilde{x}_{N-1} + \beta_1 \text{sgn}(\tilde{x}_{N-1} + \tilde{x}_f) \quad (11)$$

where  $\gamma, \beta_1 \in \mathbb{R}_{>0}$ <sup>1</sup>. When the filtered error derivative  $\dot{r}(t)$  is evaluated in the closed-loop Lyapunov analysis, the cross-terms involving the unmeasurable velocity error cancel algebraically against the embedded components of  $\dot{\eta}$  and  $v(t)$ <sup>1</sup>. This injects an effective damping term  $-kr$  into the unmeasured error dynamics, guaranteeing asymptotic state

convergence without numerical differentiation, high-gain peaking, or strict positive realness (SPR) transfer function constraints<sup>1</sup>.

### 3. Deep RNN and Physics-Informed LSTM Observers

#### Deep Recurrent Neural Network Observers

While shallow dynamic neural networks guaranteed asymptotic convergence, their single-layer formulation limited their capacity to represent multiscale, highly nonlinear vector fields without requiring a prohibitive number of parallel neurons<sup>7</sup>. To enhance representational fidelity, Griffis, Patil, Makumi, and Dixon (2023) developed a deep recurrent neural network (DRNN)-based observer for uncertain nonlinear systems<sup>10</sup>. The DRNN observer models the uncertain vector field using cascaded recurrent hidden layers<sup>8</sup>:

$$\begin{aligned}\dot{\chi}_1 &= -\Lambda_1 \chi_1 + \sigma_1(W_1 \chi_1 + U_1 y) \\ \dot{\chi}_2 &= -\Lambda_2 \chi_2 + \sigma_2(W_2 \chi_2 + V_1 \chi_1) \\ &\vdots \\ \dot{\chi}_D &= -\Lambda_D \chi_D + \sigma_D(W_D \chi_D + V_{D-1} \chi_{D-1}) \\ \hat{f} &= W_{\text{out}} \chi_D\end{aligned}\tag{12}$$

where  $\chi_d \in \mathbb{R}^{n_d}$  ( $d = 1, \dots, D$ ) represents the continuous hidden state of the  $d^{\text{th}}$  recurrent layer,  $\Lambda_d > 0$  are diagonal decay matrices,  $W_d$  denote recurrent hidden weights,  $V_d$  are inter-layer transition weights, and  $W_{\text{out}}$  is the output projection matrix<sup>10</sup>.

The primary theoretical hurdle in deep architectures is that the ideal weights appear within deeply nested, non-convex compositions of activation functions<sup>8</sup>. Griffis et al. formulated a continuous-time framework where every layer's weight matrix is adapted online via analytical Jacobians derived from Lyapunov stability analysis, bypassing the non-convex optimization traps of gradient-descent BPTT<sup>8</sup>.

#### Continuous-Time Lyapunov-Based LSTM Observers

Despite the functional richness of continuous deep RNNs, standard recurrent hidden units remain susceptible to vanishing gradients when learning dynamical effects over extended temporal horizons<sup>13</sup>. To overcome this limitation, Griffis, Patil, Hart, and Dixon (2024) formulated the Lyapunov-Based Long Short-Term Memory (Lb-LSTM) adaptive observer<sup>15</sup>.

Conventional LSTM architectures are defined in discrete time, which creates an incompatibility when coupled with continuous-time physical plants<sup>13</sup>. To resolve this, Griffis et al. transformed the discrete gating mechanics into a continuous differential framework<sup>13</sup>. The continuous-time

Lb-LSTM observer estimates unmeasured states while concurrently updating the weights across all four LSTM gates (forget, input, cell candidate, and output gates) in real time<sup>13</sup>. By updating the cell and hidden states continuously, the Lb-LSTM retains long-term dynamic memory, allowing the observer to maintain tracking accuracy through prolonged dynamic transients and sensor noise<sup>13</sup>.

## Physics-Informed LSTM (PI-LSTM) Architectures

A fundamental weakness of unconstrained black-box neural networks is their tendency to yield physically implausible state estimates during transient extrapolation or under severe noise<sup>19</sup>. In mechanical and robotic systems, physical laws dictate conservation principles and structural matrix symmetries that must hold universally across the configuration space<sup>21</sup>.

To embed physical validity into continuous neural observers, Hart, Griffis, Patil, and Dixon (2024) developed the Lyapunov-Based Physics-Informed LSTM (PI-LSTM) framework<sup>16</sup>. Focusing on general Euler-Lagrange dynamical systems, the plant is expressed as<sup>13</sup>:

$$M(q)\ddot{q} + V_m(q, \dot{q})\dot{q} + G(q) + F(\dot{q}) = u + d(t) \quad (13)$$

where  $M(q) \in \mathbb{R}^{n \times n}$  represents the generalized inertia matrix,  $V_m(q, \dot{q}) \in \mathbb{R}^{n \times n}$  is the centripetal-Coriolis matrix,  $G(q) \in \mathbb{R}^n$  is the gravitational torque vector, and  $F(\dot{q}) \in \mathbb{R}^n$  represents generalized friction<sup>13</sup>.

Rather than utilizing an unconstrained black-box LSTM to approximate the aggregate acceleration field, the PI-LSTM decomposes the network into modular, physically meaningful sub-networks that separately estimate the individual dynamic matrices: an inertia sub-network

$\hat{M}(q, \hat{\theta}_M)$ , a centripetal-Coriolis sub-network  $\hat{V}_m(q, \dot{q}, \hat{\theta}_V)$ , a gravity sub-network  $\hat{G}(q, \hat{\theta}_G)$ , and a dissipative friction sub-network  $\hat{F}(\dot{q}, \hat{\theta}_F)$ <sup>21</sup>.

The adaptive update laws are synthesized to strictly enforce foundational mechanical

properties during online learning<sup>21</sup>. First, the estimated inertia matrix  $\hat{M}(q, \hat{\theta}_M)$  is structurally parameterized using Cholesky factorizations or symmetric tensor mappings to ensure that it remains symmetric and strictly positive definite across the configuration space<sup>21</sup>:

$$\hat{M}(q, \hat{\theta}_M) = \hat{M}^T(q, \hat{\theta}_M) > 0, \quad \forall q \in \mathbb{R}^n \quad (14)$$

Second, the centripetal-Coriolis network is coupled to the inertia network such that the matrix

$\dot{\hat{M}} - 2\hat{V}_m$  preserves the fundamental skew-symmetry condition<sup>21</sup>:

$$z^T \left( \dot{\hat{M}}(q, \hat{\theta}_M) - 2\hat{V}_m(q, \dot{q}, \hat{\theta}_V) \right) z = 0, \quad \forall z \in \mathbb{R}^n \quad (15)$$

By embedding these physical invariants directly into the neural network architecture, the PI-LSTM prevents non-physical divergence, reduces the volume of the learnable parameter space, and improves generalization across unexplored regions of the system state space<sup>16</sup>.

## 4. Mathematical Framework, Continuous Discretization, and Stability Mechanisms

### Continuous-Time LSTM Formulation via Forward Euler Discretization

To apply Lyapunov analysis to recurrent gated units, the discrete update mechanics must be expressed as continuous differential equations<sup>13</sup>. Discrete-time LSTMs are conventionally formulated as:

$$\begin{aligned} f_k &= \sigma_g(W_f z_k), \quad i_k = \sigma_g(W_i z_k), \quad o_k = \sigma_g(W_o z_k), \quad c_k^* = \sigma_c(W_c z_k) \\ c_k &= f_k \odot c_{k-1} + i_k \odot c_k^* \\ h_k &= o_k \odot \sigma_c(c_k) \end{aligned} \quad (16)$$

where  $\odot$  denotes the Hadamard element-wise product,  $\sigma_g$  is the sigmoid activation function, and  $\sigma_c$  is the hyperbolic tangent activation function<sup>9</sup>.

The transition to continuous time is achieved via a forward Euler discretization limit<sup>13</sup>. Let

$\Delta t > 0$  represent an infinitesimal sampling interval. The time derivative of the continuous cell state  $c(t)$  is approximated by:

$$\dot{c}(t) \approx \frac{c_k - c_{k-1}}{\Delta t} \quad (17)$$

Substituting the discrete cell recurrence  $c_k = f_k \odot c_{k-1} + i_k \odot c_k^*$  into (17) yields:

$$\dot{c}(t) \approx \frac{f_k \odot c_{k-1} - c_{k-1} + i_k \odot c_k^*}{\Delta t} = - \left( \frac{I - f_k}{\Delta t} \right) c_{k-1} + \frac{i_k \odot c_k^*}{\Delta t} \quad (18)$$

By defining continuous decay rates and positive scaling constants  $b_c, b_h \in \mathbb{R}_{>0}$ , the continuous-time representation of the LSTM cell is formulated as a set of coupled non-autonomous ordinary differential equations<sup>13</sup>:

$$\begin{aligned} \dot{c}(t) &= -b_c c(t) + b_c \bar{c}(x, c, h, \theta) \\ \dot{h}(t) &= -b_h h(t) + b_h \bar{h}(x, c, h, \theta, W_o) \end{aligned} \quad (19)$$

where the augmented input feature vector incorporates the continuous hidden state and an explicit bias input<sup>13</sup>:

$$z(t) \triangleq [x^T(t), h^T(t), 1]^T \in \mathbb{R}^{l_1} \quad (20)$$

The continuous-time gate evaluation vectors are defined as<sup>13</sup>:

$$\begin{aligned} f(z, W_f) &= \sigma_g(W_f z) \in \mathbb{R}^{l_2} \\ i(z, W_i) &= \sigma_g(W_i z) \in \mathbb{R}^{l_2} \\ o(z, W_o) &= \sigma_g(W_o z) \in \mathbb{R}^{l_2} \\ c^*(z, W_c) &= \sigma_c(W_c z) \in \mathbb{R}^{l_2} \end{aligned} \quad (21)$$

The auxiliary functions  $\bar{c} \in \mathbb{R}^{l_2}$  and  $\bar{h} \in \mathbb{R}^{l_2}$  driving the continuous state updates are expressed as<sup>13</sup>:

$$\begin{aligned}\bar{c}(x, c, h, \theta) &\triangleq f(z, W_f) \odot c(t) + i(z, W_i) \odot c^*(z, W_c) \\ \bar{h}(x, c, h, \theta, W_o) &\triangleq o(z, W_o) \odot \sigma_c(c(t))\end{aligned}\quad (22)$$

where  $\theta \triangleq [W_c, W_i, W_f] \in \mathbb{R}^{l_2 \times 3l_1}$  contains the internal gating matrices, and  $W_o \in \mathbb{R}^{l_2 \times l_1}$  is the output gate matrix<sup>13</sup>. The parameters  $b_c, b_h > 0$  govern the bandwidth of the cell and hidden-state low-pass filtering, stabilizing the continuous recurrence against unmodeled high-frequency dynamics<sup>13</sup>.

### Derivation of Analytical Weight Jacobians

In conventional deep learning, updates rely on discrete backpropagation<sup>2</sup>. In Lyapunov-based adaptive observation, however, update laws are continuous differential equations  $\dot{\hat{\theta}} = \tau(e)$  driven by state estimation errors<sup>2</sup>. Because the LSTM output  $\Phi(x, \hat{\Theta}) = \hat{W}_v h(x, \hat{\theta}, \hat{W}_o)$  is highly nonlinear with respect to the inner weights, establishing stability requires computing the analytical Jacobian of the network output with respect to all vectorized parameters<sup>9</sup>:

$$\hat{\Theta} \triangleq \left[ \text{vec}(\hat{W}_v)^T, \text{vec}(\hat{W}_o)^T, \text{vec}(\hat{W}_c)^T, \text{vec}(\hat{W}_i)^T, \text{vec}(\hat{W}_f)^T \right]^T \in \mathbb{R}^l \quad (23)$$

The overall network output Jacobian is partitioned block-wise<sup>27</sup>:

$$\Phi'(\hat{\Theta}) \triangleq \frac{\partial \Phi(x, \hat{\Theta})}{\partial \hat{\Theta}} = \left[ \frac{\partial \Phi}{\partial \text{vec}(\hat{W}_v)}, \frac{\partial \Phi}{\partial \text{vec}(\hat{W}_o)}, \frac{\partial \Phi}{\partial \text{vec}(\hat{W}_c)}, \frac{\partial \Phi}{\partial \text{vec}(\hat{W}_i)}, \frac{\partial \Phi}{\partial \text{vec}(\hat{W}_f)} \right] \quad (24)$$

The Jacobian with respect to the outer projection weights  $\hat{W}_v \in \mathbb{R}^{n \times l_2}$  follows directly via the Kronecker product ( $\otimes$ )<sup>23</sup>:

$$\frac{\partial \Phi}{\partial \text{vec}(\hat{W}_v)} = h^T \otimes I_n \quad (25)$$

For the internal gate weight matrices  $\hat{W}_g$  ( $g \in \{o, c, i, f\}$ ), the Jacobian requires differentiating through the continuous activations using the chain rule<sup>9</sup>. Let

$z = [x^T, h^T, 1]^T$ . Using the identity  $\text{vec}(AXB) = (B^T \otimes A)\text{vec}(X)$  :

$$\frac{\partial \Phi}{\partial \text{vec}(\hat{W}_g)} = \hat{W}_v \frac{\partial h}{\partial \text{vec}(\hat{W}_g)} \quad (26)$$

For the output gate  $\hat{W}_o$ :

$$\frac{\partial h}{\partial \text{vec}(\hat{W}_o)} = \text{diag}(\sigma_c(c)) \cdot \frac{\partial o}{\partial \text{vec}(\hat{W}_o)} = \text{diag}(\sigma_c(c)) \cdot \text{diag} \left( \sigma'_g(\hat{W}_o z) \right) \quad (27)$$

where  $\sigma'_g(u) = \sigma_g(u) \odot (1 - \sigma_g(u))$  is the derivative of the sigmoid function.

For the inner cell gates ( $c^*, i, f$ ), the chain rule propagates through the hyperbolic tangent of the cell state  $\sigma_c(c)$ <sup>13</sup>:

$$\frac{\partial h}{\partial c} = \text{diag}(o) \cdot \text{diag} (1 - \sigma_c^2(c)) \quad (28)$$

Applying this relation to each respective gate yields:

$$\begin{aligned}
\frac{\partial \Phi}{\partial \text{vec}(\hat{W}_c)} &= \hat{W}_v \frac{\partial h}{\partial c} \cdot \text{diag}(i) \cdot \text{diag} \left( 1 - \sigma_c^2(\hat{W}_c z) \right) \cdot (z^T \otimes I_{l_2}) \\
\frac{\partial \Phi}{\partial \text{vec}(\hat{W}_i)} &= \hat{W}_v \frac{\partial h}{\partial c} \cdot \text{diag}(c^*) \cdot \text{diag} \left( \sigma'_g(\hat{W}_i z) \right) \cdot (z^T \otimes I_{l_2}) \\
\frac{\partial \Phi}{\partial \text{vec}(\hat{W}_f)} &= \hat{W}_v \frac{\partial h}{\partial c} \cdot \text{diag}(c) \cdot \text{diag} \left( \sigma'_g(\hat{W}_f z) \right) \cdot (z^T \otimes I_{l_2})
\end{aligned} \tag{29}$$

Using these expressions, the analytical Jacobian  $\Phi'(\hat{\Theta})$  is evaluated in real time without backpropagating across prior temporal horizons<sup>9</sup>. This formulation provides the sensitivity matrix required to design adaptive laws that cancel unmodeled dynamics within the Lyapunov derivative<sup>14</sup>.

### Nonsmooth Lyapunov Stability and Discontinuous Inclusions

Because the observer architectures incorporate discontinuous switching terms (such as sliding-mode signum functions  $\text{sgn}(\cdot)$  or smooth projection operators to enforce parameter compactness)<sup>1</sup>, the closed-loop error vector field is discontinuous:

$$\dot{\zeta}(t) = \mathcal{F}(\zeta(t)), \quad \zeta \triangleq [r^T, \eta^T, \tilde{x}_{N-1}^T, \tilde{x}_f^T, \tilde{\Theta}^T]^T \tag{30}$$

where  $\mathcal{F}$  is measurable but discontinuous on switching manifolds<sup>1</sup>. As a result, classical Cauchy-Lipschitz uniqueness theorems fail<sup>29</sup>. To establish mathematical rigor, the system is analyzed as a differential inclusion in the sense of Filippov<sup>2</sup>:

$$\dot{\zeta}(t) \in \mathcal{K}[\mathcal{F}](\zeta(t)) \triangleq \bigcap_{\delta > 0} \bigcap_{\mu(\mathcal{N})=0} \overline{\text{co}} \{ \mathcal{F}(B(\zeta, \delta) \setminus \mathcal{N}) \} \tag{31}$$

where  $\mathcal{K}[\cdot]$  is the Filippov regularization operator,  $\overline{\text{co}}$  denotes the closed convex hull,  $B(\zeta, \delta)$  is an open ball of radius  $\delta$  centered at  $\zeta$ , and  $\mu(\mathcal{N})$  is the Lebesgue measure of set  $\mathcal{N}$ <sup>2</sup>.

Consider a continuously differentiable, positive-definite Lyapunov function candidate  $V : \mathbb{R}^m \rightarrow \mathbb{R}_{\geq 0}$ :

$$V(\zeta) = \frac{1}{2}r^2 + \frac{1}{2}\eta^2 + \frac{1}{2}\tilde{x}_{N-1}^2 + \frac{1}{2}\tilde{x}_f^2 + \frac{1}{2}\tilde{\Theta}^T \Gamma^{-1} \tilde{\Theta} \quad (32)$$

where  $\tilde{\Theta} \triangleq \Theta^* - \hat{\Theta}$  represents the parameter estimation error,  $\Theta^*$  denotes the ideal bounding weights on a compact domain, and  $\Gamma = \Gamma^T > 0$  is an adaptive gain matrix<sup>2</sup>. Because  $V$  is smooth, its generalized time derivative along the Filippov differential inclusion satisfies Clarke's gradient<sup>2</sup>:

$$\dot{V}(\zeta) \stackrel{a.e.}{\in} \dot{\hat{V}}(\zeta) \triangleq \bigcap_{\xi \in \partial V(\zeta)} \xi^T \mathcal{K}[\mathcal{F}](\zeta) \quad (33)$$

Substituting the closed-loop error dynamics (including the auxiliary dynamic filter cancellation terms and the neural Jacobian  $\Phi'$ ) into  $\dot{\hat{V}}$  results in:

$$\dot{V} \leq -k_r r^2 - k_\eta \eta^2 - k_x \tilde{x}_{N-1}^2 - k_f \tilde{x}_f^2 + \tilde{\Theta}^T \left( \Gamma^{-1} \dot{\tilde{\Theta}} + \Phi'^T r \right) + [|r| \bar{D}] \quad (34)$$

The parameter adaptation law is defined analytically to eliminate the parametric error cross-term<sup>2</sup>:

$$\dot{\tilde{\Theta}} = \text{proj}(\Gamma \Phi'^T r) \implies \tilde{\Theta}^T \left( \Gamma^{-1} \dot{\tilde{\Theta}} + \Phi'^T r \right) \leq 0 \quad (35)$$

where  $\text{proj}(\cdot)$  is a continuous projection operator keeping  $\hat{\Theta}$  confined within a known convex compact set<sup>2</sup>. Choosing the robust feedback gain  $\beta_1 \geq \bar{D}$ , where  $\bar{D}$  upper-bounds the network reconstruction error and exogenous disturbance, ensures that the sliding term dominates the residual uncertainty<sup>1</sup>:

$$|r|\bar{D} - \beta_1 r \mathcal{K}[\text{sgn}(r)] \leq 0 \quad (36)$$

Consequently, the generalized time derivative is negative semi-definite<sup>1</sup>:

$$\dot{V}(\zeta) \leq -W(\zeta) \leq -c_1 \|\tilde{x}\|^2 \quad (37)$$

where  $W(\zeta)$  is a continuous positive semi-definite function of the state estimation errors<sup>29</sup>.

### Error Convergence via the Nonsmooth LaSalle-Yoshizawa Corollary

Because the generalized derivative in (37) is negative semi-definite (as parameter estimation errors  $\tilde{\Theta}$  reside in  $V$  but do not appear in  $W(\zeta)$ ), classical asymptotic stability theorems do not directly apply<sup>19</sup>. Furthermore, because the vector field is discontinuous, classical smooth corollaries like Barbalat's lemma cannot be invoked<sup>19</sup>.

Instead, the nonsmooth extension of the LaSalle-Yoshizawa corollary is applied<sup>19</sup>. Under this formulation, since  $V(\zeta)$  is positive definite, radially unbounded, and satisfies

$\sup \dot{V}(\zeta) \leq -W(\zeta) \leq 0$ , every Filippov solution  $\zeta(t)$  remains bounded for all  $t \geq 0$ ,

and the limit of  $W(\zeta(t))$  converges to zero as  $t \rightarrow \infty$ <sup>2</sup>. Because  $W(\zeta) = 0$  implies

that  $r = 0, \eta = 0, \tilde{x}_{N-1} = 0$ , and  $\tilde{x}_f = 0$ , all state estimation errors converge asymptotically to zero<sup>1</sup>:

$$\lim_{t \rightarrow \infty} \|\tilde{x}(t)\| = 0 \quad (38)$$

A key insight of this nonsmooth LaSalle-Yoshizawa formulation is that asymptotic convergence

of the state estimation error  $\|\tilde{x}(t)\| \rightarrow 0$  does not require a persistence of excitation (PE) condition, nor does it require offline pre-training data<sup>14</sup>. If a PE condition is satisfied, the

parameter estimation error  $\|\tilde{\Theta}(t)\|$  also converges to zero, achieving exact system identification<sup>14</sup>. However, even in the absence of PE, the observer guarantees asymptotic state reconstruction while parameter estimates remain uniformly bounded<sup>2</sup>.

## 5. Applications to Underactuated and Coupled Mechanical Systems

## Benchmark Mechanical Dynamics

The efficacy of continuous-time Lyapunov-based recurrent observers is established across two representative mechanical benchmarks: the fully coupled two-link planar robot manipulator and the underactuated, non-minimum phase cart-inverted pendulum<sup>1</sup>.

The two-link planar robot manipulator operates in the horizontal plane and is described by coupled Euler-Lagrange equations of motion<sup>1</sup>:

$$M(q)\ddot{q} + V_m(q, \dot{q})\dot{q} + G(q) + F_d\dot{q} + F_s(\dot{q}) = \tau(t) + d(t) \quad (39)$$

where  $q = [q_1, q_2]^T$  denotes link angular positions, and  $\tau = [\tau_1, \tau_2]^T$  are applied joint control torques<sup>1</sup>. The configuration-dependent inertia matrix  $M(q)$  and centripetal-Coriolis matrix  $V_m(q, \dot{q})$  are given by:

$$M(q) = \begin{bmatrix} p_1 + 2p_3 \cos(q_2) & p_2 + p_3 \cos(q_2) \\ p_2 + p_3 \cos(q_2) & p_2 \end{bmatrix}, \quad V_m(q, \dot{q}) = \begin{bmatrix} -p_3 \dot{q}_2 \sin(q_2) \\ p_3 \dot{q}_1 \sin(q_2) \end{bmatrix} \quad (40)$$

where  $p_1, p_2, p_3$  are physical link inertial parameters<sup>1</sup>. The terms  $F_d\dot{q}$  and  $F_s(\dot{q})$  introduce viscous and nonlinear Stribeck friction<sup>26</sup>:

$$F_s(\dot{q}) = \begin{bmatrix} \gamma_1 (\tanh(\beta_{s1}\dot{q}_1) - \tanh(\beta_{s2}\dot{q}_1)) \\ \gamma_2 (\tanh(\beta_{s3}\dot{q}_2) - \tanh(\beta_{s4}\dot{q}_2)) \end{bmatrix} \quad (41)$$

Only link positions  $q_1, q_2$  are measurable; joint velocities  $\dot{q}_1, \dot{q}_2$  must be reconstructed in the presence of unmodeled friction and exogenous torque disturbances<sup>1</sup>.

The cart-inverted pendulum represents an underactuated, non-minimum phase mechanical benchmark with generalized coordinate vector  $q = [x, \theta]^T$ , where  $x$  denotes the horizontal cart displacement and  $\theta$  is the pendulum angle relative to the upright vertical equilibrium<sup>19</sup>:

$$\begin{aligned}
(M_c + m)\ddot{x} + ml\ddot{\theta}\cos(\theta) - ml\dot{\theta}^2\sin(\theta) + b_c\dot{x} &= u \\
ml\ddot{x}\cos(\theta) + (J + ml^2)\ddot{\theta} - mgl\sin(\theta) + b_p\dot{\theta} &= 0
\end{aligned} \tag{42}$$

where  $u$  represents the single control force applied to the cart,  $M_c$  is cart mass,  $m$  is pendulum mass,  $l$  is the pendulum length to the center of mass,  $J$  is the moment of inertia, and  $b_c, b_p$  represent damping and pivot friction coefficients<sup>19</sup>. The system is underactuated because the single control input  $u$  must stabilize both the actuated horizontal cart position and the unactuated rotational pendulum angle<sup>19</sup>. Encoders provide discrete position measurements of  $x$  and  $\theta$ , while translational velocity  $\dot{x}$  and angular velocity  $\dot{\theta}$  must be reconstructed despite sensor noise and dry friction<sup>1</sup>.

### Comparative Estimator Evaluation

To benchmark continuous-time Lyapunov-based recurrent observers against classical estimation paradigms, their performance is evaluated across three conventional state estimation techniques<sup>1</sup>:

First, numerical dirty derivatives approximate velocities by passing measured position signals  $y(t)$  through a causal high-pass lead filter:

$$\hat{y}(s) = \frac{s}{\tau_d s + 1} y(s) \implies \tau_d \ddot{\hat{y}}(t) + \dot{\hat{y}}(t) = \dot{y}(t) \tag{43}$$

While computationally straightforward, selecting the filter time constant  $\tau_d$  involves an inherent trade-off<sup>1</sup>. A small  $\tau_d$  introduces minimal phase lag but significantly amplifies high-frequency measurement noise<sup>1</sup>. A larger  $\tau_d$  attenuates high-frequency noise but introduces substantial phase delay, which erodes stability margins and often destabilizes underactuated systems like the inverted pendulum<sup>1</sup>.

Second, the Extended Kalman Filter (EKF) approximates state estimates by propagating a continuous-discrete Riccati equation based on localized first-order Jacobian linearizations<sup>3</sup>:

$$F_k \approx \left. \frac{\partial f(x, u)}{\partial x} \right|_{\hat{x}_{k|k-1}}, \quad P_{k|k-1} = F_k P_{k-1|k-1} F_k^T + Q_k \quad (44)$$

The EKF assumes Gaussian process noise  $Q_k$  and measurement noise  $R_k$ <sup>3</sup>. In the presence of nonlinear friction regimes (such as Stribeck effects and stick-slip transitions in (41)), localized linearizations can miscalculate covariance matrices, resulting in filter divergence or large steady-state tracking offsets<sup>3</sup>. Furthermore, the EKF does not provide deterministic Lyapunov stability guarantees under structural model uncertainties<sup>3</sup>.

Third, High-Gain Observers (HGO) reconstruct unmeasured states for systems in canonical form by scaling the observer gains with an inverted small parameter  $\epsilon \ll 1$ <sup>1</sup>:

$$\dot{\hat{x}}_1 = \hat{x}_2 + \frac{\alpha_1}{\epsilon}(y - \hat{x}_1), \quad \dot{\hat{x}}_2 = f_0(\hat{x}, u) + \frac{\alpha_2}{\epsilon^2}(y - \hat{x}_1) \quad (45)$$

where  $f_0$  is a nominal model. As  $\epsilon \rightarrow 0$ , the observer error converges arbitrarily fast to an  $O(\epsilon)$  boundary layer, effectively rejecting modeling errors<sup>1</sup>. However, HGOs suffer from the peaking phenomenon: during initial transients, estimation errors exhibit spikes of order  $O(1/\epsilon)$ <sup>5</sup>. When saturated actuators or closed-loop feedback are engaged, this transient peak can destabilize the physical plant. Moreover, high gains amplify high-frequency measurement noise by  $1/\epsilon^2$ , resulting in severe chattering in the estimated velocity states<sup>1</sup>.

| Operational Metric       | Numerical Dirty Derivatives                                                      | Extended Kalman Filter (EKF)                                                    | High-Gain Observer (HGO)                                                        | Lb-LSTM / PI-LSTM Observer                                                     |
|--------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Noise Sensitivity</b> | Highly sensitive; amplifies sensor noise proportional to $1/\tau_d$<br>[cite: 1] | Moderate; filters Gaussian noise effectively via covariance tuning <sup>3</sup> | Severe; amplifies measurement noise proportional to $1/\epsilon^N$<br>[cite: 1] | Low; internal low-pass cell dynamics filter high-frequency noise <sup>13</sup> |

|                               |                                                                                        |                                                                                    |                                                                                             |                                                                                      |
|-------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Transient Peaking</b>      | No internal dynamic peaking, but exhibits large initial phase lag <sup>1</sup>         | Minimal peaking under appropriate initial covariance initialization <sup>3</sup>   | Severe peaking phenomenon ( $O(1/\epsilon)$ magnitude spikes) <sup>5</sup>                  | Peaking eliminated via Lyapunov-bounded weight projection <sup>2</sup>               |
| <b>Robustness to Friction</b> | Invariant to model friction, but velocity estimates lag stick-slip events <sup>1</sup> | Degrades under unmodeled Stribeck friction; linearization breaks down <sup>3</sup> | Rejects friction errors at the cost of high noise amplification <sup>1</sup>                | Identifies and cancels nonlinear friction online via memory cells <sup>8</sup>       |
| <b>Stability Guarantees</b>   | None; open-loop differentiation without stability proof <sup>1</sup>                   | Local asymptotic stability under restrictive linearizability bounds <sup>3</sup>   | Semi-global practical asymptotic stability <sup>5</sup>                                     | Deterministic asymptotic/exponential convergence via Lyapunov proofs <sup>1</sup>    |
| <b>Underactuated Tracking</b> | Destabilizes underactuated dynamics due to phase lag and noise injection <sup>1</sup>  | Prone to divergent oscillations around non-minimum phase points <sup>3</sup>       | Can stabilize underactuated modes, but peaking risks exceeding actuator limits <sup>5</sup> | Reconstructs unmeasured coordinates accurately without phase lag <sup>15</sup>       |
| <b>Computational Overhead</b> | Minimal; single-pole causal digital filter <sup>1</sup>                                | Moderate; continuous matrix inversion and covariance propagation <sup>3</sup>      | Minimal; fixed linear error gain injection <sup>1</sup>                                     | High; real-time evaluation of analytical Jacobians across gating layers <sup>9</sup> |

In experimental and simulation studies conducted across the Dixon literature, the integration of continuous-time Lb-LSTM and auxiliary filter designs achieves a 40% to 70% reduction in function approximation and velocity estimation errors relative to classical baselines<sup>20</sup>. Unlike HGOs, the Lb-LSTM does not rely on high gains to reject uncertainty, which prevents noise peaking and limits control effort<sup>1</sup>. Unlike EKF, it maintains stable state reconstruction despite large unmodeled friction and discontinuous dynamics<sup>3</sup>.

## 6. Critical Synthesis and Future Horizons

The research lineage pioneered by Prof. Warren E. Dixon's NCR laboratory systematically resolved the central limitation of nonlinear adaptive observers: how to maintain deterministic Lyapunov stability guarantees when employing complex, non-convex machine learning models without relying on offline pre-training data<sup>1</sup>.

Beginning with Dinh et al. (2014), the formulation of the second-order auxiliary dynamic filter established that unmeasurable velocity error dependencies can be cancelled algebraically, enabling asymptotic convergence of state estimation errors via sliding-mode injection without dirty derivatives<sup>1</sup>. Subsequent work advanced this paradigm from shallow dynamic neural networks to deep recurrent architectures and continuous-time LSTMs<sup>10</sup>. By converting discrete gating mechanics into continuous differential equations via forward Euler approximations, continuous-time Lb-LSTMs leverage temporal memory to capture long-term dynamic dependencies without suffering from vanishing gradients<sup>13</sup>. The derivation of analytical Jacobians across all inner and outer gating weights made it possible to bypass offline BPTT, enabling stable, real-time weight adaptation<sup>9</sup>. Coupled with Filippov differential inclusions and the nonsmooth LaSalle-Yoshizawa corollary, these architectures guarantee asymptotic error convergence under bounded disturbances without requiring persistent excitation<sup>2</sup>.

Recent research at the intersection of control theory and machine learning points toward several emerging extensions of this framework. One active branch explores Lyapunov-Based Kolmogorov-Arnold Networks (Lb-KAN), which replace standard multilayer perceptron layers with learnable univariate spline activations along the network edges<sup>19</sup>. By decoupling multivariate approximations into visualizable, additive single-variable functions, Lb-KAN observers offer structural interpretability while reducing functional approximation errors by roughly 20% compared to baseline Lb-LSTM architectures<sup>19</sup>.

Another significant trajectory involves concurrent learning without persistence of excitation<sup>8</sup>. Conventional adaptive observers require restrictive, infinite-horizon persistence of excitation (PE) conditions on the system trajectories to guarantee parameter convergence (exact system identification)<sup>14</sup>. By maintaining an online history stack of recorded, linearly independent input-output data, concurrent learning adaptation laws achieve exponential parameter convergence once a finite-time rank condition is satisfied, enabling simultaneous state estimation and true physical parameter identification along standard non-exciting operational paths<sup>8</sup>.

Finally, the continuous neural observer framework is increasingly integrated with safety-critical control methodologies through Adaptive Deep Neural Network Control Barrier Functions (aDCBFs)<sup>27</sup>. By combining continuous Lb-LSTM estimators with control barrier functions, the convergent bounds on parameter and state estimation errors can be mapped directly into safe set margins within real-time quadratic programs<sup>27</sup>. This synthesis guarantees forward invariance and system safety even in the presence of unmodeled nonlinear dynamics, sensor noise, or intermittent loss of state feedback<sup>27</sup>.

## Works cited

1. Dynamic Neural Network-based Robust Observers for Uncertain, <https://scc-lab.github.io/Preprints/SCC.Dinh.Kamalapurkar.ea2014.pdf>
2. Distributed State Estimation With Deep Neural Networks for, <https://ncr.mae.ufl.edu/papers/tac233.pdf>
3. Simultaneous Estimation of Euclidean Distances to a Stationary, <https://ncr.mae.ufl.edu/papers/tac2021.pdf>
4. Asymptotic Tracking for Uncertain Dynamic Systems via a Multilayer, [https://ncr.mae.ufl.edu/papers/tac08\\_2.pdf](https://ncr.mae.ufl.edu/papers/tac08_2.pdf)
5. Publications - Nonlinear Controls and Robotics - University of Florida, <https://ncr.mae.ufl.edu/aacoe.php?id=aa/publications>
6. Distributed Coordination of Multiple Unknown Euler-Lagrange, <https://ncr.mae.ufl.edu/papers/TCNS18c.pdf>
7. Exploring Flip Flop memories and beyond: training Recurrent Neural, <https://www.frontiersin.org/journals/systems-neuroscience/articles/10.3389/fnsys.2024.1269190/full>
8. Omkar PATIL | University of Florida, Gainesville | UF | Research profile, <https://www.researchgate.net/profile/Omkar-Patil-5>
9. Stability Analysis of Soft Sensor-Based Control Systems--A Lur'e, [https://www.researchgate.net/publication/391878616\\_Stability\\_Analysis\\_of\\_Soft\\_Sensor-Based\\_Control\\_Systems--A\\_Lur'e-Postnikov\\_Approach\\_Using\\_a\\_Novel\\_Gated\\_Recurrent\\_Neural\\_Network](https://www.researchgate.net/publication/391878616_Stability_Analysis_of_Soft_Sensor-Based_Control_Systems--A_Lur'e-Postnikov_Approach_Using_a_Novel_Gated_Recurrent_Neural_Network)
10. Emily Griffis - Google Scholar, [https://scholar.google.com/citations?user=Krd\\_E80AAAAJ&hl=en](https://scholar.google.com/citations?user=Krd_E80AAAAJ&hl=en)
11. Publications - Nonlinear Controls and Robotics - University of Florida, <https://ncr.mae.ufl.edu/index.php?id=publications>
12. Warren Dixon - Nonlinear Controls and Robotics - University of Florida, <https://ncr.mae.ufl.edu/index.php?id=people>
13. Lyapunov-Based Long Short-Term Memory (Lb-LSTM) Neural, <https://ncr.mae.ufl.edu/papers/csl234.pdf>
14. Rebecca G. Hart - Google Scholar, <https://scholar.google.com/citations?user=KS17-5kAAAAJ&hl=en>
15. Warren E. Dixon - DBLP, <https://dblp.org/pid/51/208>
16. Warren E. Dixon | OpenReview, [https://openreview.net/profile?id=~Warren\\_E.\\_Dixon1](https://openreview.net/profile?id=~Warren_E._Dixon1)
17. Huyen Dinh - Google Scholar, <https://scholar.google.com/citations?user=y-hqtrsAAAAJ&hl=en>
18. NCR Affiliated Publications for Rebecca Hart, <https://ncr.mae.ufl.edu/index.php?id=people/BeccaH>
19. Deep Residual Neural Network (ResNet)-Based Adaptive Control, [https://www.researchgate.net/publication/367032533\\_Deep\\_Residual\\_Neural\\_Network\\_ResNet-Based\\_Adaptive\\_Control\\_A\\_Lyapunov-Based\\_Approach](https://www.researchgate.net/publication/367032533_Deep_Residual_Neural_Network_ResNet-Based_Adaptive_Control_A_Lyapunov-Based_Approach)
20. Neural Map: Structured Memory for Deep Reinforcement Learning, [https://www.researchgate.net/publication/314092651\\_Neural\\_Map\\_Structured\\_M](https://www.researchgate.net/publication/314092651_Neural_Map_Structured_M)

[emory\\_for\\_Deep\\_Reinforcement\\_Learning](#)

21. Lyapunov-Based Physics-Informed Long Short-Term Memory (LSTM),  
[https://www.researchgate.net/publication/376840509\\_Lyapunov-Based\\_Physics-Informed\\_Long\\_Short-Term\\_Memory\\_LSTM\\_Neural\\_Network-Based\\_Adaptive\\_Control](https://www.researchgate.net/publication/376840509_Lyapunov-Based_Physics-Informed_Long_Short-Term_Memory_LSTM_Neural_Network-Based_Adaptive_Control)
22. (PDF) Lyapunov-Based Physics-Informed Deep Neural Networks,  
[https://www.researchgate.net/publication/396924620\\_Lyapunov-Based\\_Physics-Informed\\_Deep\\_Neural\\_Networks\\_with\\_Skew\\_Symmetry\\_Considerations](https://www.researchgate.net/publication/396924620_Lyapunov-Based_Physics-Informed_Deep_Neural_Networks_with_Skew_Symmetry_Considerations)
23. Collaborative Spacecraft Servicing Under Partial Feedback Using,  
<https://ncr.mae.ufl.edu/papers/JAS25.pdf>
24. [https://en.wikipedia.org/wiki/Long\\_short-term\\_memory](https://en.wikipedia.org/wiki/Long_short-term_memory)
25. PUBLICATIONS CONTENT DIGEST - IEEE Control Systems Society,  
[https://ieeecss.org/sites/ieeecss/files/documents/pcd/css\\_publications\\_content\\_digest\\_022024.pdf](https://ieeecss.org/sites/ieeecss/files/documents/pcd/css_publications_content_digest_022024.pdf)
26. Tracking Control in the Presence of Nonlinear Dynamic Frictional,  
<https://ncr.mae.ufl.edu/papers/ajc99.pdf>
27. (PDF) Adaptive Deep Neural Network-Based Control Barrier Functions,  
[https://www.researchgate.net/publication/381604577\\_Adaptive\\_Deep\\_Neural\\_Network-Based\\_Control\\_Barrier\\_Functions](https://www.researchgate.net/publication/381604577_Adaptive_Deep_Neural_Network-Based_Control_Barrier_Functions)
28. The Robust Control of a Nonsmooth or Switched Control-Affine,  
<https://www.mdpi.com/2076-3417/15/8/4482>
29. Theory - Nonlinear Controls and Robotics - University of Florida,  
<https://ncr.mae.ufl.edu/index.php?id=research/theory>
30. Switching of Asymptotically Stable and Uniformly Ultimately,  
[https://ncr.mae.ufl.edu/papers/tac24\\_5.pdf](https://ncr.mae.ufl.edu/papers/tac24_5.pdf)